

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 2

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Nissanki Sri Venkata Lakshmi Kavya(192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
N.S.V.L.Kavya

1. Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

A)Event Communication Security – Caesar Cipher (Shift = 4)

Question:

A student organization wants to securely share the message "MEET AT NOON" using a Caesar Cipher with a fixed shift of 4 positions.

- a) Encrypt the message.
- b) Explain how the receiver gets back the original message.

Introduction

The Caesar Cipher is one of the simplest classical encryption techniques. Each letter in the message is shifted by a fixed number of positions in the alphabet. Here, every letter is shifted 4 positions forward.

Step 1: Encryption (Shift = 4)

Original Letter	Encrypted Letter
M	Q
E	I
E	I
T	X
A	E
T	X
N	R
O	S
O	S
N	R

Original Message: MEET AT NOON

Encrypted Message: QIIX EX RSSR

B)How the Encryption Works

M → Q

E → I

$E \rightarrow I$
 $T \rightarrow X$

$A \rightarrow E$
 $T \rightarrow X$

$N \rightarrow R$
 $O \rightarrow S$
 $O \rightarrow S$
 $N \rightarrow R$

Step 2: Decryption

The authorized member knows that the sender used a shift of 4. To recover the original message, each encrypted letter is shifted 4 positions backward.

$Q \rightarrow M$
 $I \rightarrow E$
 $I \rightarrow E$
 $X \rightarrow T$
 $E \rightarrow A$
 $X \rightarrow T$
 $R \rightarrow N$
 $S \rightarrow O$
 $S \rightarrow O$
 $R \rightarrow N$

Decrypted Message: MEET AT NOON

2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.

b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

a. Deduce the most likely original message.

The given message is:

Cipher Text:

GSRH RH Z HVXIVG

This message is encrypted using the Atbash Cipher.

In the Atbash Cipher, each letter is replaced by its opposite letter in the alphabet.

Alphabet Mapping

Cipher Letter	Original Letter
G	T
S	H
R	I
H	S
Z	A
V	E
X	C
I	R

Original Message

THIS IS A SECRET

b. Explain how you identified the original message.

The message was identified using the Atbash Cipher because it follows a simple letter replacement pattern.

- In the Atbash Cipher, each letter is replaced with its opposite letter in the alphabet.
- For example:
 - A becomes Z
 - B becomes Y
 - C becomes X
 - D becomes W

After replacing every letter in "GSRH RH Z HVXIVG", the message becomes:

THIS IS A SECRET

The words "THIS", "IS", and "SECRET" are meaningful English words. Since the entire sentence makes sense and follows the Atbash Cipher rule, we can conclude that the original message is:

THIS IS A SECRET

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation